# Spatiotemporal Prediction of Electric Vehicle Charging Demand

Data Science Lab Project Report

Submitted by: _____

Roll number: _____

Course: Data Science Lab

Guide: _____

Institution: _____

Date: _____

*Dataset: UrbanEV - Li, H. et al. (2025), Scientific Data 12, 523. All figures and numbers in this document were generated from the project's executed code.*

## 1. Introduction

Electric vehicles (EVs) are spreading quickly in large cities, and charging infrastructure has to keep up. Shenzhen, China, has one of the largest electric fleets in the world, including taxis and ride-hailing vehicles that charge many times a day. How much energy is drawn from chargers changes strongly from hour to hour and from one part of the city to another.

This project builds a machine-learning system that forecasts the hourly EV charging demand (in kWh) of every charging zone in Shenzhen one hour ahead. Three regression models (Linear Regression, Random Forest and LightGBM) are trained on the public UrbanEV dataset and compared on a strict chronological test period. The best model is deployed in a local web application (Streamlit) where users can view forecasts per zone, on a city map and over a full day.

## 2. Problem Statement

Charging demand varies greatly by time and location: some stations are under-used while others are overcrowded at peak hours. This affects drivers (queues), charging operators (capacity and pricing), urban planners (where to add chargers) and electricity providers (grid load). The task is to **predict the total energy (kWh) delivered by EV chargers in each zone during the next hour**, using only information available before that hour starts. Because the target is a continuous number, this is a **supervised regression problem**.

## 3. Objectives

- Collect, inspect and understand the UrbanEV spatiotemporal charging dataset for Shenzhen.
- Preprocess and integrate the charging, zone, station, charger-power and adjacency data.
- Engineer leakage-free features for next-hour forecasting.
- Analyse relationships between demand and the other variables using Pearson correlation.
- Train Linear Regression, Random Forest and LightGBM regressors and compare them with MAE, MSE, RMSE and $R^2$.
- Interpret the models (coefficients, feature importance, residuals) and select the best one.
- Deploy the selected model in a user-friendly web application.

## 4. Dataset Description

**Source:** UrbanEV - An Open Benchmark Dataset for Urban Electric Vehicle Charging Demand Prediction (Li et al., Scientific Data, 2025). The zone-level, cleaned and aggregated version is used, with hourly resolution. The files are provided as CSV.

| Property | Value |
|---|---|
| Charging zones | 275 |
| Hourly timestamps | 4,344 |
| Period | 2022-09-01 to 2023-02-28 |
| Zone-hour records | 1,194,600 |
| Charging stations (raw station file) | 1,682 |
| Charging piles with rated power | 22,650 (19,556 AC, 3,094 DC) |
| Missing values in hourly files | 0 |

### 4.1 Files used

| File | Content | Shape |
|---|---|---|
| charge_1hour/volume.csv | Energy delivered per zone-hour (kWh) - TARGET | 4,344 x 275 (wide) |
| charge_1hour/occupancy.csv | Number of charging piles in use | 4,344 x 275 |
| charge_1hour/duration.csv | Charging duration | 4,344 x 275 |
| charge_1hour/e_price.csv | Electricity price (RMB/kWh) | 4,344 x 275 |
| charge_1hour/s_price.csv | Service price (RMB/kWh) | 4,344 x 275 |
| charge_1hour/volume-11kw.csv | Energy from <=11 kW chargers (part of volume) | 4,344 x 275 |
| zone-information.csv | Zone ID, longitude, latitude, charger count, area, perimeter | 275 x 6 |
| station_information.csv (raw) | Station -> zone, slow / fast charger counts | 1,682 x 7 |
| pile_rated_power.csv | Charger (pile) power in kW and AC/DC type | 22,650 x 4 |
| adj.csv / distance.csv | Zone adjacency (0/1) and distances (m) | 275 x 275 |
| charge_5min/volume.csv | 5-minute volume (used only for a consistency check) | 52,128 x 275 |

Each hourly file is a **wide** table: one time column plus one column per zone ID. The zone columns, the TAZID values in zone-information.csv and the row/column order of adj.csv and distance.csv were verified to refer to the same 275 zones in the same order.

**Target variable volume:** mean 261.5 kWh per zone-hour, median 53.7 kWh, maximum 16,732.5 kWh; 5.5% of zone-hours have zero demand.

## 5. Data Preprocessing

- **Reshaping:** the six wide hourly files were stacked into one long table with one row per (time, zone): 1,194,600 rows.
- **Missing values, duplicates and negative values:** none were found (0 in every file). Forward-fill, duplicate removal and clipping at 0 are kept as safeguards.
- **Time consistency:** 4,344 consecutive hours with no gaps or duplicate timestamps.
- **Data types:** measurements converted to float32 and zone IDs to integers to save memory.
- **Invalid coordinates:** zone 348 had longitude/latitude = 0; it was replaced by the mean location of the zone's stations.
- **Adjacency matrix:** 20 one-directional links were made symmetric; 4 zones without neighbours use their 3 nearest zones (from distance.csv).
- **Scaling:** only Linear Regression needs it; a StandardScaler is fitted on training data only, inside a pipeline.

### 5.1 Data validation checks

| Check | Result | Conclusion |
|---|---|---|
| Hourly volume = sum of 5-minute volumes | max difference 0.000434 kWh | Hourly data is consistent |
| Zone charger counts = sum of station slow + fast counts | 100% match | Station file used for charger features |
| e_price daily profile repeats previous day | 97.4% of zone-days | Prices are a known tariff |
| s_price daily profile repeats previous day | 97.1% of zone-days | Prices are a known tariff |
| volume_11kw <= volume | 99.4% of rows | Component of target - excluded |
| Adjacent zones' mean distance | about 2 km (vs 22 km non-adjacent) | Adjacency aligned correctly |

### 5.2 Data integration

Static attributes were joined to each zone-hour by zone ID: number of chargers, share of fast chargers, average rated power per charger (from pile_rated_power.csv, linked via station_id -> zone) and total charger power (capacity, kW). Hourly demand never exceeds the zone's total charger power, which confirms the capacity feature is physically consistent.

## 6. Feature Engineering

**Prediction horizon: next hour.** A row (zone, hour t) has the target volume at hour t. All features use information from hour t-1 or earlier, or information known in advance for hour t (calendar, public holidays, tariff prices, zone attributes). Lags are computed separately for each zone.

| Group | Features | Meaning |
|---|---|---|
| Demand history | volume_lag1, volume_lag24, volume_lag168, volume_roll24_mean | Zone's own demand 1 h, 24 h and 168 h earlier; mean of the last 24 h |
| Activity history | occupancy_lag1, duration_lag1 | Chargers in use and charging duration in the previous hour |
| Spatial | neighbor_volume_lag1 | Average previous-hour demand of adjacent zones |
| Tariff | e_price, s_price | Electricity and service price for the hour (published tariff) |
| Calendar | hour, day, dayofweek, is_weekend, is_holiday, hour_sin, hour_cos, dow_sin, dow_cos | Hour, day, weekday, weekend and holiday flags, sine/cosine cycles |
| Zone | num_piles, fast_share, avg_pile_kw, capacity_kw, area, longitude, latitude | Charger count and power, zone area and location |

In total **25 input features** are used. month is created but not used as an input: training covers Sep-Dec while the test set is February, a value never seen in training. The first 168 hours are dropped because the weekly lag is not yet defined.

### 6.1 Chronological split

| Split | Rows | Start | End |
|---|---|---|---|
| train | 759,000 | 08 Sep 2022 | 31 Dec 2022 |
| val | 204,600 | 01 Jan 2023 | 31 Jan 2023 |
| test | 184,800 | 01 Feb 2023 | 28 Feb 2023 |

All zones of one timestamp are always in the same split, and the splits never overlap in time. The validation month is used only for LightGBM early stopping and for model selection.

### 6.2 Leakage prevention

- Same-hour occupancy and duration are only known after the hour ends, so only their 1-hour lags are used.
- volume_11kw is part of the target and is never used as an input.
- Lag values were spot-checked on 2,000 random rows against the original data (all correct).
- Scaling and every learnt quantity are fitted on training data only.

# 7. Correlation Analysis

Exploratory analysis is limited to Pearson correlation, computed on the training period (759,000 rows). Same-hour occupancy, duration and volume_11kw are included only for discussion.

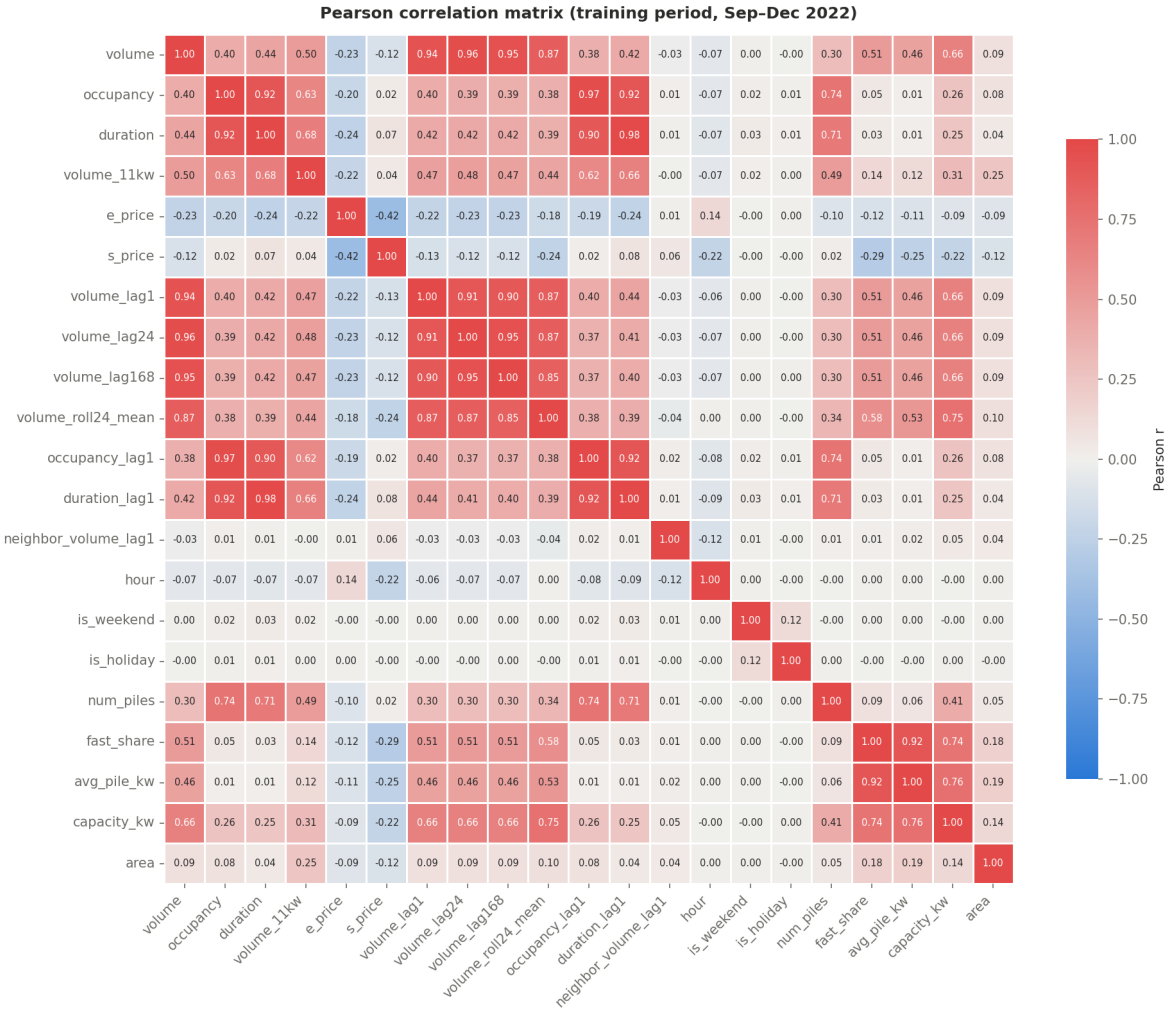



*Figure 1. Pearson correlation matrix (training period).*

## 7.1 Correlation with the target

| Feature | Pearson r | Strength | Role |
|---|---|---|---|
| volume_lag24 | 0.96 | strong positive | model input |
| volume_lag168 | 0.95 | strong positive | model input |
| volume_lag1 | 0.94 | strong positive | model input |
| volume_roll24_mean | 0.87 | strong positive | model input |
| capacity_kw | 0.66 | moderate positive | model input |
| fast_share | 0.51 | moderate positive | model input |
| volume_11kw | 0.50 | moderate positive | excluded - part of the target (leakage) |
| avg_pile_kw | 0.46 | moderate positive | model input |
| duration | 0.44 | moderate positive | excluded - same-hour value unknown at forecast time |
| duration_lag1 | 0.42 | moderate positive | model input |
| occupancy | 0.40 | moderate positive | excluded - same-hour value unknown at forecast time |
| occupancy_lag1 | 0.38 | weak positive | model input |

| Feature | Pearson r | Strength | Role |
|---|---|---|---|
| num_piles | 0.30 | weak positive | model input |
| e_price | -0.23 | weak negative | model input |
| s_price | -0.12 | negligible | model input |
| area | 0.09 | negligible | model input |
| hour | -0.07 | negligible | model input |
| neighbor_volume_lag1 | -0.03 | negligible | model input |
| is_weekend | 0.00 | negligible | model input |
| is_holiday | -0.00 | negligible | model input |

## 7.2 Pooled vs within-zone correlation

| Feature | Pooled r | Median within-zone r |
|---|---|---|
| volume_lag1 | 0.94 | 0.93 |
| volume_lag24 | 0.96 | 0.56 |
| volume_lag168 | 0.95 | 0.39 |
| volume_roll24_mean | 0.87 | 0.49 |
| occupancy_lag1 | 0.38 | 0.67 |
| duration_lag1 | 0.42 | 0.93 |
| neighbor_volume_lag1 | -0.03 | 0.30 |
| e_price | -0.23 | -0.40 |
| s_price | -0.12 | 0.29 |
| hour_cos | 0.05 | 0.28 |

## 7.3 Multicollinearity (|r| >= 0.8 among inputs)

| Feature 1 | Feature 2 | Pearson r |
|---|---|---|
| volume_lag24 | volume_lag168 | 0.95 |
| occupancy_lag1 | duration_lag1 | 0.92 |
| fast_share | avg_pile_kw | 0.92 |
| volume_lag1 | volume_lag24 | 0.91 |
| volume_lag1 | volume_lag168 | 0.90 |
| volume_lag1 | volume_roll24_mean | 0.87 |
| volume_lag24 | volume_roll24_mean | 0.87 |
| volume_lag168 | volume_roll24_mean | 0.85 |

## 7.4 Interpretation

- Strongest predictors among model inputs: volume_lag24 (r = 0.96), volume_lag168 (r = 0.95), volume_lag1 (r = 0.94). Recent demand in the same zone is the best single indicator of next-hour demand.
- Same-hour occupancy (r = 0.40) and duration (r = 0.44) move with demand but are only known after the hour ends, so the models use their 1-hour lags instead (r = 0.38 and 0.42).
- volume_11kw has r = 0.50 with the target because it is a component of it (energy from <=11 kW chargers); using it would leak the answer.
- Electricity price r = -0.23 and service price r = -0.12: prices follow a daily time-of-use tariff, so part of their correlation reflects the time of day rather than a direct price effect.
- Zone size/capacity: num_piles r = 0.30, capacity_kw r = 0.66 - larger zones deliver more energy on average.
- hour has r = -0.07: Pearson only measures LINEAR association, and the daily cycle is not linear in

the hour number, which is why sine/cosine hour features are used.

- neighbor_volume_lag1 has pooled r = -0.03, but inside each zone over time the median r is 0.30. Pooled correlation is dominated by size differences between zones (a large zone can border small ones); within a zone, busier neighbours do go with busier hours.
- hour_cos has median within-zone r = 0.28, confirming a clear daily cycle that the raw hour number hides.
- Negative correlations: e_price (r = -0.23)
- Multicollinearity (|r| >= 0.8) among inputs: volume_lag24 ~ volume_lag168 (r = 0.95); occupancy_lag1 ~ duration_lag1 (r = 0.92); fast_share ~ avg_pile_kw (r = 0.92); volume_lag1 ~ volume_lag24 (r = 0.91); volume_lag1 ~ volume_lag168 (r = 0.90); volume_lag1 ~ volume_roll24_mean (r = 0.87); volume_lag24 ~ volume_roll24_mean (r = 0.87); volume_lag168 ~ volume_roll24_mean (r = 0.85). This makes Linear Regression coefficients unstable to interpret, but does not harm tree-based models' predictions.
- Correlation does not imply causation; it only measures linear co-movement.

# 8. Algorithms Used

## 8.1 Linear Regression

Models the target as a weighted sum of the features: ŷ = b0 + b1·x1 + ... + bn·xn. The weights minimise the sum of squared errors (ordinary least squares). It is fast and interpretable, but it can only represent linear relationships, and its coefficients become unstable when features are strongly correlated (multicollinearity). Features are standardised so coefficients are comparable.

## 8.2 Random Forest Regressor

An ensemble of decision trees (bagging). Each tree is trained on a random sample of rows and considers a random subset of features at each split; the forest prediction is the average of all trees. Averaging reduces variance, captures non-linear relationships and interactions, and gives impurity-based feature importance.

## 8.3 LightGBM Regressor

A gradient-boosting framework. Trees are built one after another; each new tree is fitted to the remaining errors (gradients) of the current ensemble, and its contribution is scaled by a learning rate. LightGBM grows trees leaf-wise and uses histogram-based splitting, which makes it fast and memory-efficient on large tabular data. Early stopping on the validation set prevents overfitting.

## 8.4 Naive reference forecasts

Two forecasts that involve no learning are reported for context: repeat the previous hour's demand (lag-1) and repeat the demand at the same hour one week earlier (lag-168). A useful model must clearly beat them.

# 9. Model Implementation

| Model | Configuration |
|---|---|
| Linear Regression | StandardScaler -> LinearRegression (scikit-learn) |
| Random Forest | 100 trees, max_depth 18, min_samples_leaf 20, max_features 0.5, max_samples 0.3 |
| LightGBM | learning_rate 0.05, num_leaves 63, min_child_samples 50, subsample 0.8, colsample_bytree 0.8, up to 3,000 trees with early stopping (100 rounds) on validation RMSE |
| All models | same 25 features, same training rows, random seed 42, predictions clipped at 0 kWh |

| Model | Training rows | Training time (s) | Saved size (MB) |
|---|---|---|---|
| Linear Regression | 759,000 | 0.90 | 0.00 |
| Random Forest | 759,000 | 52.30 | 23.60 |
| LightGBM | 759,000 | 14.90 | 0.80 |

LightGBM stopped early at 369 trees.

## 9.1 Software architecture

- **Backend (src/):** data loading, preprocessing, feature engineering, correlation analysis, training, evaluation and a prediction service. python train.py runs the whole pipeline once.
- **Frontend (Streamlit app):** Home, Demand Forecast, Demand Map, Zone Insights, Compare Zones and About the Model pages. The app only loads the saved best model and calls the prediction service; no training happens in the interface.
- **Notebook:** the full analysis, using the same src modules so results are identical.
- **Persistence:** models saved with joblib; data cached as Parquet; results saved as CSV/JSON/PNG.

# 10. Evaluation Metrics

- **MAE** = $(1/n)\ \Sigma\ |y - \hat{y}|$ - the average absolute error in kWh; easy to interpret.
- **MSE** = $(1/n)\ \Sigma\ (y - \hat{y})^2$ - squares errors, so large errors are penalised strongly (units kWh²).
- **RMSE** = $\sqrt{MSE}$ - like MSE but back in kWh; used for model selection.
- **$R^2$** = $1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$ - share of the variance in demand explained by the model (1 = perfect).

All metrics are computed on the February 2023 test period, which was never used for training or model selection.

# 11. Results and Comparison

| Model | MAE (kWh) | MSE | RMSE (kWh) | $R^2$ |
|---|---|---|---|---|
| Linear Regression | 48.70 | 33,799.61 | 183.85 | 0.9535 |
| Random Forest | 35.56 | 26,215.02 | 161.91 | 0.9639 |
| LightGBM | 34.28 | 20,355.55 | 142.67 | 0.9720 |
| Naive (lag-1) | 51.49 | 82,381.59 | 287.02 | 0.8867 |
| Naive (lag-168) | 77.19 | 104,835.52 | 323.78 | 0.8558 |

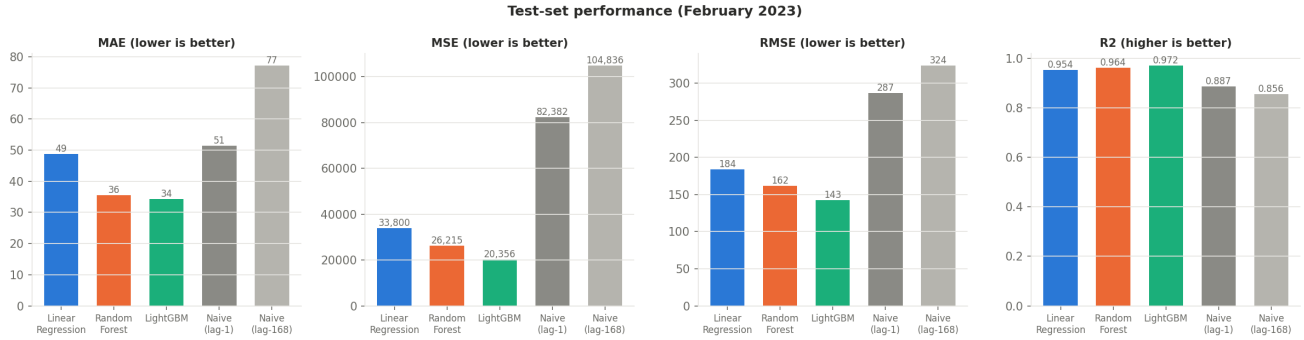



*Figure 2. Test-set metrics for all models and baselines.*

## 11.1 Overfitting and underfitting check

| Model | Train RMSE | Validation RMSE | Test RMSE |
|---|---|---|---|
| Linear Regression | 190.91 | 176.15 | 183.85 |
| Random Forest | 157.93 | 152.18 | 161.91 |
| LightGBM | 123.47 | 141.37 | 142.67 |

The tree models show small gaps between training and test error (LightGBM: 123.5 -> 142.7 kWh), so there is little overfitting. Linear Regression has a higher error even on its own training data (190.9 kWh), a sign of underfitting: the relationship is not purely linear.

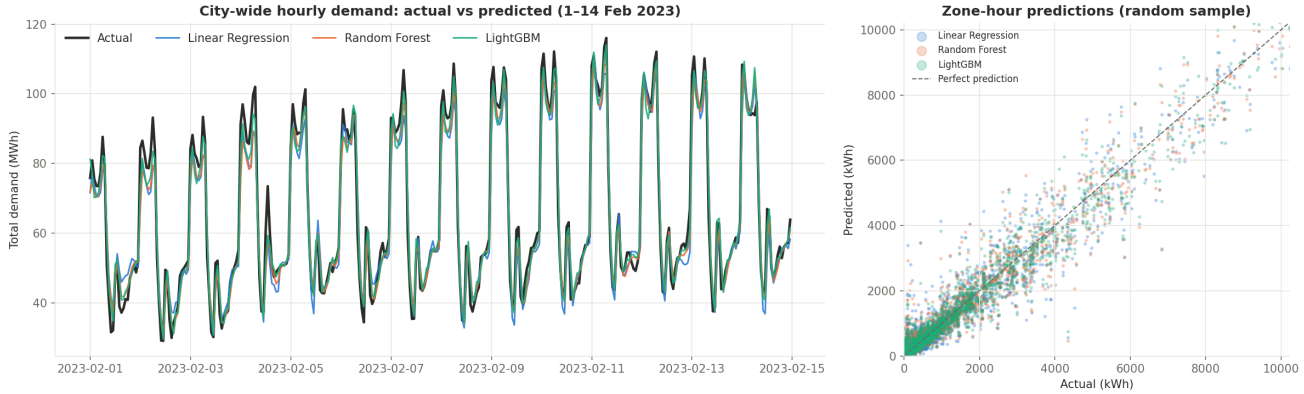



*Figure 3. Actual vs predicted demand on the test period.*

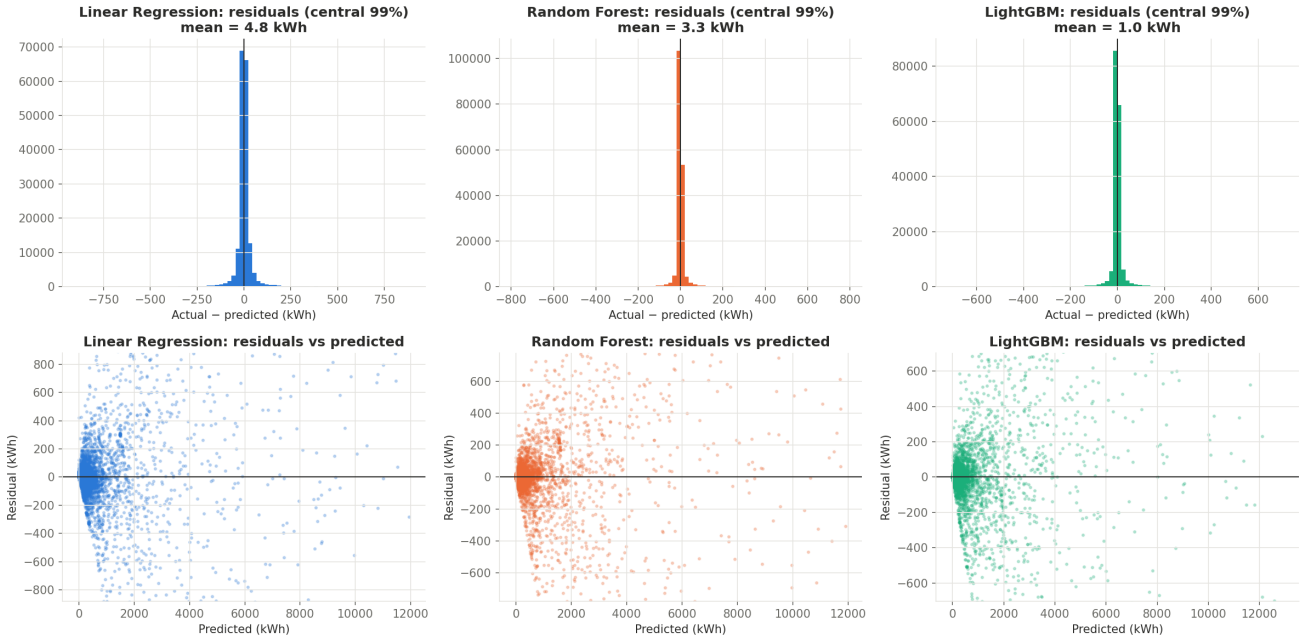



*Figure 4. Residual analysis (actual - predicted).*

Residuals are centred close to zero for all models. Errors grow for zones with high demand, which is expected because large zones vary more in absolute kWh.

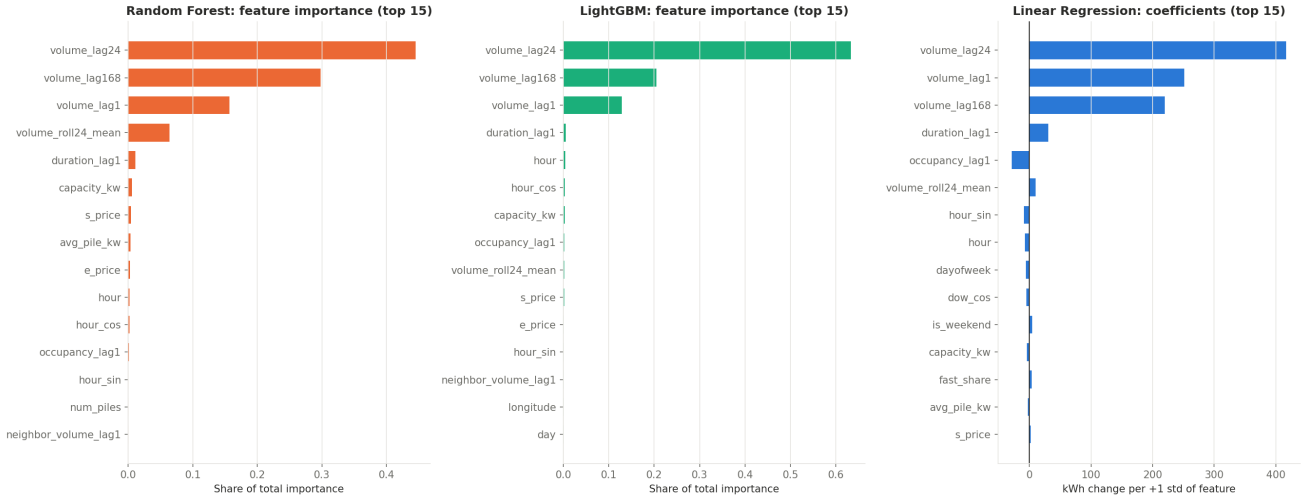



*Figure 5. Feature importance (tree models) and Linear Regression coefficients.*

## 11.2 Linear Regression coefficients (top 10)

| Feature | Coefficient (kWh per +1 std) |
|---|---|
| volume_lag24 | 417.21 |
| volume_lag1 | 251.44 |
| volume_lag168 | 219.78 |
| duration_lag1 | 30.67 |
| occupancy_lag1 | -28.31 |
| volume_roll24_mean | 10.20 |
| hour_sin | -9.13 |
| hour | -7.46 |
| dayofweek | -5.97 |
| dow_cos | -4.73 |

### 11.3 LightGBM feature importance (top 10)

| Feature | Share of importance |
|---|---|
| volume_lag24 | 63.2% |
| volume_lag168 | 20.5% |
| volume_lag1 | 12.9% |
| duration_lag1 | 0.6% |
| hour | 0.5% |
| hour_cos | 0.3% |
| capacity_kw | 0.3% |
| occupancy_lag1 | 0.2% |
| volume_roll24_mean | 0.2% |
| s_price | 0.2% |

### 11.4 Final model selection

The model with the lowest validation RMSE is **LightGBM** (Linear Regression: 176.2, Random Forest: 152.2, LightGBM: 141.4). The choice was confirmed on the test set, where LightGBM achieved MAE 34.3 kWh, RMSE 142.7 kWh and $R^2$ 0.9720. RMSE reduction: 22.4% vs Linear Regression, 11.9% vs Random Forest, 50.3% vs Naive (lag-1).

Gradient boosting builds trees sequentially, each correcting the remaining errors, so it captures non-linear effects (daily cycles, zone size, interactions between recent and weekly demand) that a linear equation cannot, and with early stopping it avoids overfitting. Random Forest averages independent trees and is limited in depth for memory reasons, which leaves slightly more error.

## 12. Conclusion

- The UrbanEV dataset was fully inspected and validated (1,194,600 zone-hours); all files and identifiers align.
- A leakage-free next-hour forecasting pipeline with 25 features was built and verified.
- All three models clearly outperform the naive forecasts; **LightGBM** is the best model ($R^2$ 0.972, average error 34.3 kWh per zone-hour).
- The most influential features are volume_lag24, volume_lag168, volume_lag1: recent demand history, especially the same hour on the previous day, drives next-hour demand.
- Demand is concentrated overnight: 46% of all energy is delivered between 00:00 and 07:59, when electricity prices are lowest.
- Zones with more fast chargers and higher installed power deliver more energy.
- The selected model is deployed in a working local web application for end users.

**Practical applications:** operators can plan capacity, maintenance and dynamic pricing; drivers can avoid busy hours; planners can identify zones where chargers are under pressure; grid operators can anticipate EV charging load.

## 13. Limitations

- Only six months of data: yearly seasonality cannot be learnt, and calendar month cannot be used as a feature.
- Results are specific to Shenzhen and its large electric taxi fleet; other cities may behave differently.
- No weather or event data; public holidays are represented only by a flag.
- Forecasts are one hour ahead and evaluated on historical data; the application is not connected to live data.
- Small inconsistencies between source files (e.g. the rated-power list covers fewer stations than the station file) were documented, not corrected.
- Random Forest was size-limited to run on a laptop, which may cost some accuracy.

## 14. Future Scope

- Add weather, traffic and event data, and a full public-holiday calendar.
- Forecast several hours or days ahead (multi-step forecasting).
- Compare with sequence and graph models (LSTM/GRU, STGCN, Graph WaveNet) that use the adjacency matrix directly.
- Connect the application to live charging data for real-time forecasts and alerts.
- Apply the pipeline to other cities' charging data to test generalisation.

## 15. References


- Li, H., Qu, H., Tan, X., You, L., Zhu, R., & Fan, W. (2025). UrbanEV: An Open Benchmark Dataset for Urban Electric Vehicle Charging Demand Prediction. Scientific Data, 12, 523. https://doi.org/10.1038/s41597-025-04874-4
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